Sanity Checks for Saliency Maps.

Core Problem: Visual Inspection is poor guide. sensitivity of explanations

Prediction based on noise or model parameters. only for images.

$X \in \mathbb{R}^d$

$S : \mathbb{R}^d \to \mathbb{R}^C$

map that assigns relevance to each dimension $E : \mathbb{R}^d \to \mathbb{R}^d$

$E_{grad}(x) = \frac{\partial S}{\partial x}$

different types of saliency maps

$E_{grad \odot input}(x) = x \odot \frac{\partial S}{\partial x}$

IG ..

$E_{IG}(x) = (x - \bar{x}) \times \int_0^1 \frac{\partial S(\bar{x} + \alpha(x - \bar{x}))}{\partial x} \partial \alpha$

$\bar{x} \to 0$ black

Guided back propagation $f^l = \max(f^{l-1}, 0)$: forward activation.

$R^{l+1} = \frac{\partial f^{out}}{\partial f^{l+1}}$

$R^l = 1_{f^{l+1} > 0} \, 1_{R^l > 0} \, R^{l+1}$

Smoothgrad ... ..

Sanity checks:

Model-Parameter .. Randomization Test.

Trained vs untrained,

Data Randomization. labelled vs randomly labelled model

$y = 50x_1 + 2x_2$

$S = 1x_1 + 1x_2 + 1x_3 \ldots$

Similarity ~~metrics~~ metrics

Cascading Randomization. layer wise destruction.

top to bottom → output → .. → CNN

weights are used or not

deepest layer most imp (only for CNN)

Independent layer randomization → only one layer.

Model parameters. ✗ not dependent the saliency maps.

guided Backprop → edge detector.

Gradient and Grad Cam ✓.

data randomization – $\boxed{\text{label}}$ change →

still smooth grad, grad ✓ Guided Backprop ✗.

$$F_{grad}(x) = \frac{\partial(w \cdot x)}{\partial x} = w$$

$$T_{SG}(x) = w.$$

$$E_{IG}(x) = (x - \bar{x}) \times \int_0^1 \frac{\partial (w \cdot (\bar{x} + \alpha(x - \bar{x})))}{\partial x} d\alpha = (x - \bar{x}) \odot w/2$$

1-layer CNN setup.. Theoretical validation

RELU pointwise σ

$$l(x) \quad \sum_{i=1}^{n} \sum_{j=1}^{n} \sigma(w * x)_{ij}$$

σ → RELU

$$\frac{\partial}{\partial x_{ij}} l(x) \quad \sum_{k=-1}^{1} \sum_{l=-1}^{1} \sigma'((w * x)_{i+k, j+l}) \, w_{kl}$$

$\boxed{3 \times 3}$ weight.

$$\sum_{k=-1}^{1} \sum_{l=-1}^{1} a_{i+k, j+l} \, w_{kl}.$$

Gradient ✓ Grad Cam.

Test ✓

Sketch → method fail → edge detector